

Objective:

To demonstrate ArrayStoreException

Execution steps:

STEP 1:

Open IntelliJ IDEA.

STEP 2:

Create a Java project.

STEP 3:

Create a package named:

javacore

STEP 4:

Create a Java class named:

ArrayStoreException

STEP 5:

Write the given program in the class.

STEP 6:

Save the program.

STEP 7:

Compile the program.

STEP 8:

Run the program.

STEP 9:

Execution starts from the main() method.

STEP 10:

The statement

```
Object[] arr = new String[2];
```

creates a String array of size 2.

STEP 11:

The try block starts executing.

STEP 12:

"Hello" is stored at index 0.

STEP 13:

The program tries to store integer 10 at index 1.

STEP 14:

An Integer cannot be stored in a String array.

STEP 15:

ArrayStoreException occurs.

STEP 16:

The catch block handles the exception.

STEP 17:

"Cannot store Integer in String array" is displayed.

STEP 18:

"Program ended" is displayed.

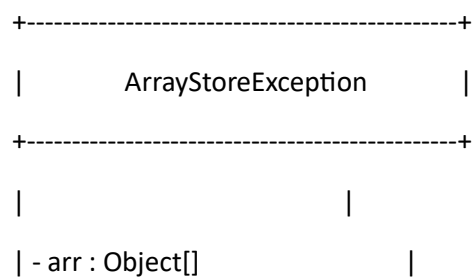
STEP 19:

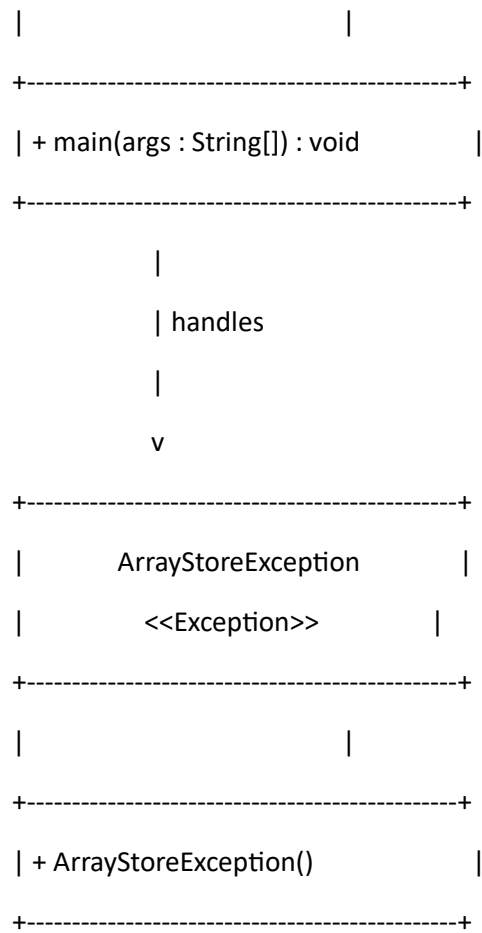
The program ends.

Algorithm:

1. Start the program.
2. Create an Object array with size 2,
where the actual array type is String.
3. Start the try block.
4. Store "Hello" in the first array position.
5. Try to store the integer 10 in the second array position.
6. Since the array can store only String values,
ArrayStoreException occurs.
7. Catch the exception using the catch block.
8. Display "Cannot store Integer in String array".
9. Display "Program ended".
10. Stop the program.

UML Class Diagram:





Code:

```
package javacore;
```

```
public class ArrayStoreException {
    public static void main(String[] args) {
```

```
        Object[] arr = new String[2];
```

```
        try {
            arr[0] = "Hello";
            arr[1] = 10;
        }
```

```
        catch (ArrayStoreException e) {
            System.out.println("Cannot store Integer in String array");
```

```
}
```

```
    System.out.println("Program ended");
```

```
}
```

```
}
```

Output:

Cannot store Integer in String array

Program ended